CHAPTER FOUR

RESULTS

4.1 Introduction

This chapter presents the analysis and interpretation of data collected to examine the effect of gasoline exposure duration on the expression of three biomarkers: Interleukin 1 beta (IL1β), Tumor Necrosis Factor alpha (TNF-α), and Caspase-3. The data were analyzed using the Statistical Package for the Social Sciences (SPSS) version 20, and the results are presented using descriptive and inferential statistical methods.

Descriptive statistics such as mean, standard deviation, and confidence intervals were used to summarize the expression profiles across the six exposure durations (control, 1 hour, 3 hours, 5 hours, 7 hours, and 9 hours). A one-way Analysis of Variance (ANOVA) was then conducted for each biomarker to determine whether exposure time had a significant effect on expression levels. Where significant differences occurred, Tukey's post hoc test identified the specific groups responsible for the variation. The results are presented in tables and figures, including bar charts with 95% confidence intervals to visualize expression patterns.

4.2 Descriptive Statistics

This section presents the descriptive statistics of the expression levels of the three biomarkers across varying exposure durations to gasoline fumes. The analysis shows how the mean expression and standard deviation change over time, providing insight into the temporal response pattern of each biomarker. The descriptive results were summarized using means and standard deviations, while bar charts were used to visualize the expression trends with 95% confidence intervals.

4.2.1 Interleukin 1 Beta (IL1β)

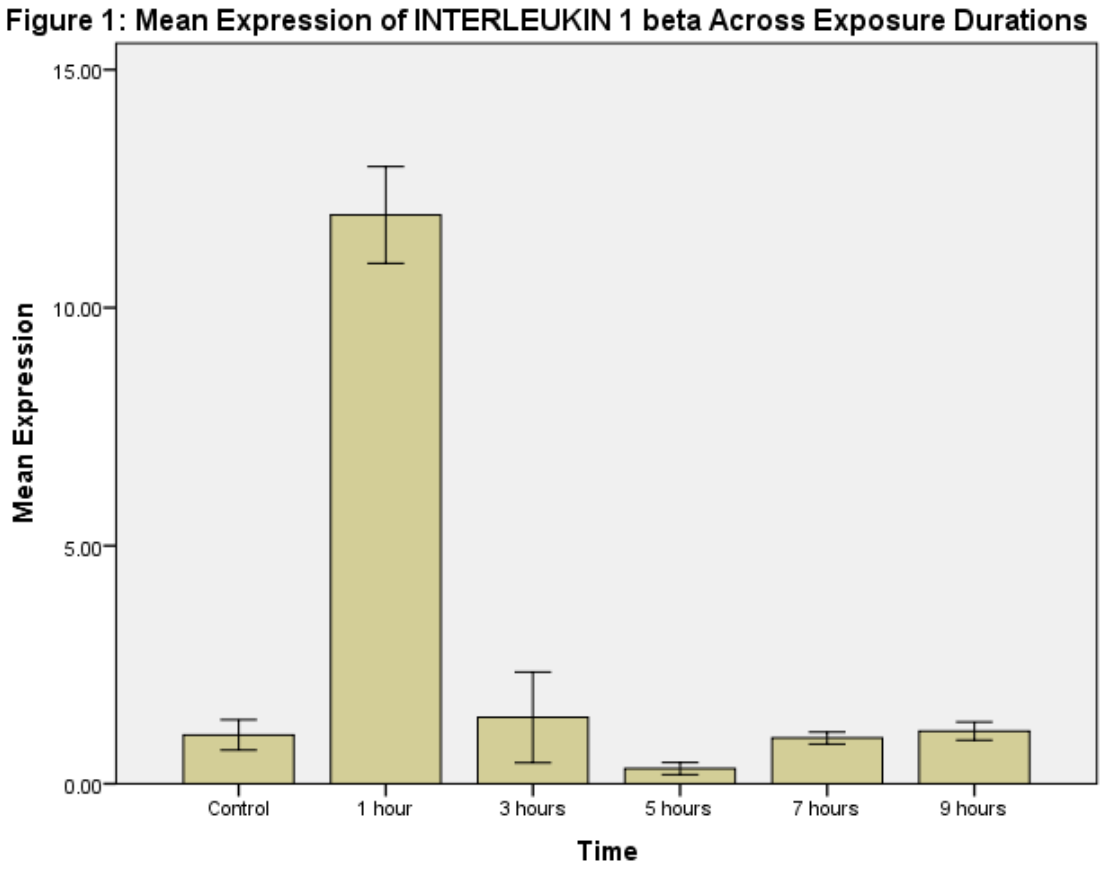
Table 4.1 presents the descriptive statistics of Interleukin 1 Beta (IL1β) expression levels across the six exposure durations. The mean expression of IL1β was highest after 1 hour of exposure (11.95 ± 0.11), representing a sharp increase from the control value of 1.03 ± 0.04. Expression dropped substantially after 3 hours (1.40 ± 0.11) and reached its lowest level at 5 hours (0.32 ± 0.01). From the 7th to the 9th hour, the expression levels stabilized close to baseline (0.96 ± 0.01 and 1.11 ± 0.02, respectively).

This pattern indicates a rapid and transient induction of IL1β following exposure to gasoline fumes. The early spike in expression after 1 hour suggests a quick inflammatory response, which declined as the exposure continued, implying that IL1β is an early-phase biomarker of inflammation. The near-normal levels observed from 7 to 9 hours indicate recovery or adaptation to the exposure over time.

Table 4.1: Descriptive Statistics of IL1β Expression Across Exposure Durations

Exposure Duration	N	Mean	Std. Deviation
Control	2	1.03	0.04
1 Hour	2	11.95	0.11
3 Hours	2	1.40	0.11
5 Hours	2	0.32	0.01
7 Hours	2	0.96	0.01
9 Hours	2	1.11	0.02

Figure 4.1: *Mean Expression of IL1β Across Exposure Durations*



4.2.2 Tumor Necrosis Factor Alpha (TNF-α)

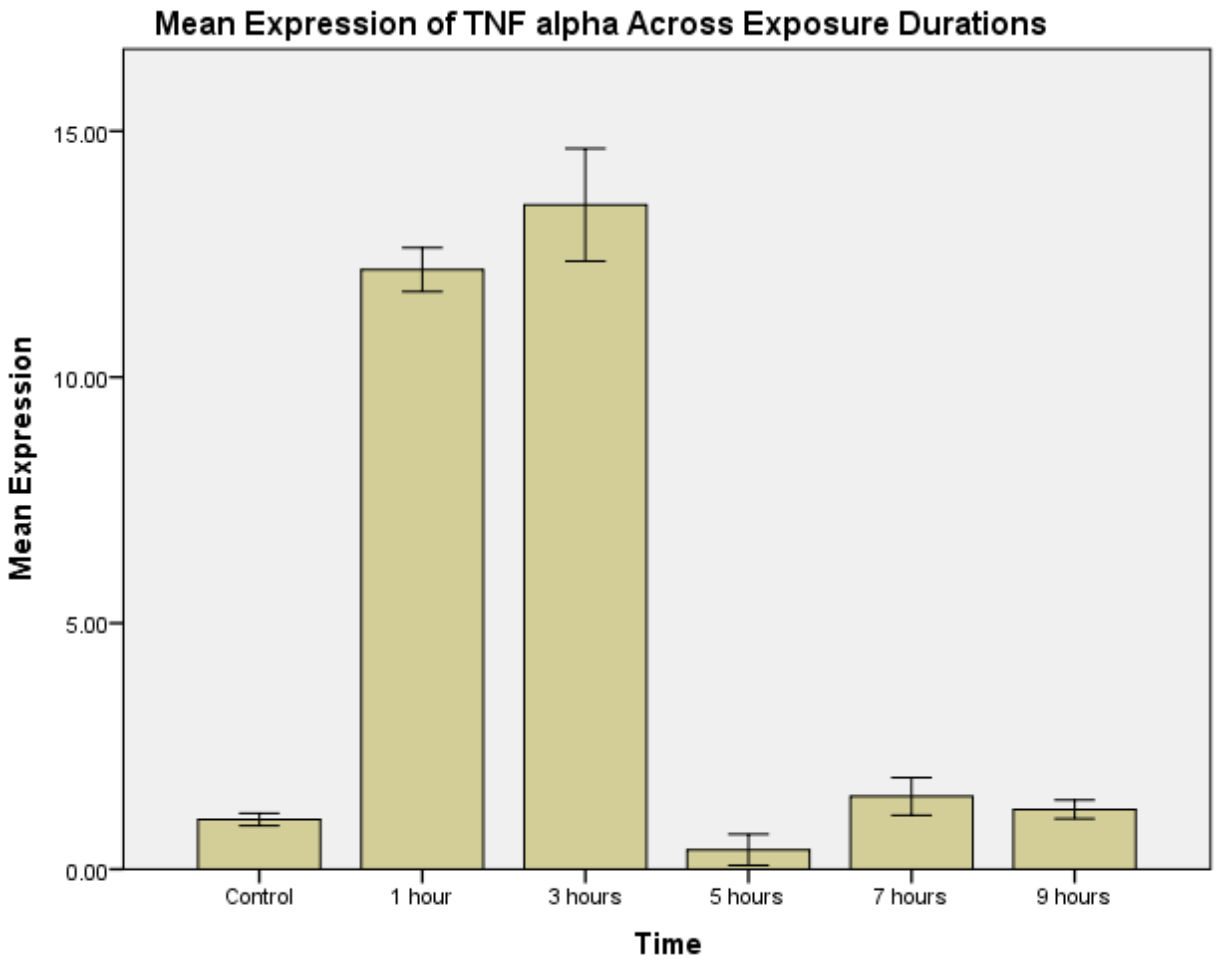
Table 4.2 summarizes the descriptive statistics of TNF-α expression levels across the six exposure durations. The analysis shows notable differences among the time groups, with the mean expression increasing sharply at the early stages of exposure. The highest mean value of 13.50 ± 0.13 was recorded after 3 hours, followed closely by 12.19 ± 0.05 at 1 hour, while the control recorded the lowest expression level (1.01 ± 0.01). A marked reduction occurred at 5 hours (0.40 ± 0.04), after which expression slightly increased again at 7 hours (1.48 ± 0.04) and 9 hours (1.22 ± 0.02).

This pattern indicates that TNF-α expression responds rapidly to gasoline exposure, peaking within the first few hours and then declining substantially as exposure continues. The initial elevation represents an early inflammatory reaction, while the later stabilization near baseline suggests a self-regulatory feedback mechanism limiting prolonged inflammation. The expression trend for TNF-α is illustrated in Figure 4.2, which displays the mean ± 95 % confidence intervals across all exposure times.

Table 4.2: Descriptive Statistics of TNF-α Expression Across Exposure Durations

Exposure Duration	N	Mean	Std. Deviation
Control	2	1.01	0.01
1 Hour	2	12.19	0.05
3 Hours	2	13.50	0.13
5 Hours	2	0.40	0.04
7 Hours	2	1.48	0.04
9 Hours	2	1.22	0.02

Figure 4.2: *Mean Expression of TNF-α Across Exposure Durations*



4.2.3 Caspase-3

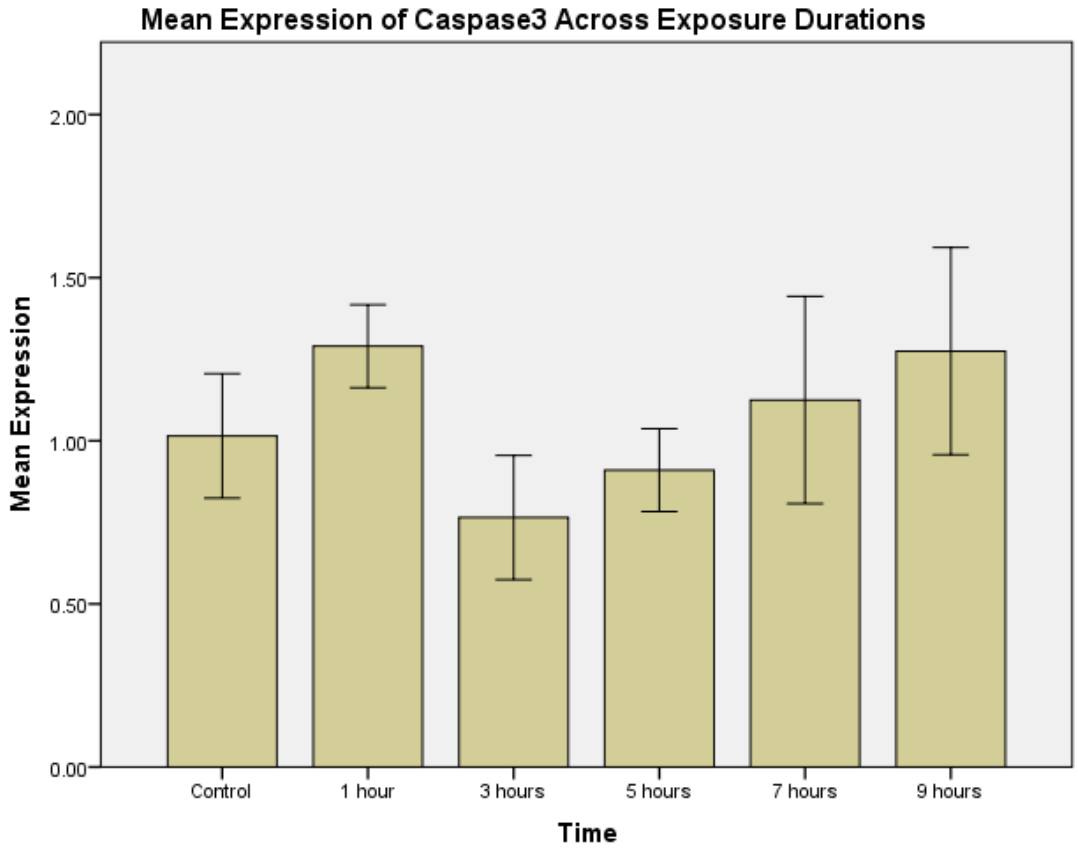
Table 4.3 presents the descriptive statistics of Caspase-3 expression across the six exposure durations. The results show a mild fluctuation in expression levels following gasoline exposure. The highest mean value was recorded at 1 hour (1.29 ± 0.01), followed closely by 9 hours (1.28 ± 0.04) and 7 hours (1.13 ± 0.04). The control group maintained a near-baseline expression of 1.02 ± 0.02, while the lowest mean was observed at 3 hours (0.77 ± 0.02), followed by a slight increase at 5 hours (0.91 ± 0.01).

This trend suggests that Caspase-3 activation exhibited a delayed but sustained response pattern compared to the other biomarkers. The early increase at 1 hour may reflect the initiation of apoptotic signaling in response to gasoline exposure, while the dip at 3 hours indicates transient suppression before reactivation at later hours (7 and 9 hours). This pattern implies that apoptotic mechanisms became more pronounced as exposure continued, which aligns with the typical role of Caspase-3 as a downstream effector in programmed cell death pathways. The expression pattern for Caspase-3 is further illustrated in Figure 4.3, which shows mean expression values with 95% confidence intervals across all time points.

Table 4.3: Descriptive Statistics of Caspase-3 Expression Across Exposure Durations

Exposure Duration	N	Mean	Std. Deviation
Control	2	1.02	0.02
1 Hour	2	1.29	0.01
3 Hours	2	0.77	0.02
5 Hours	2	0.91	0.01
7 Hours	2	1.13	0.04
9 Hours	2	1.28	0.04

Figure 4.3: *Mean Expression of Caspase-3 Across Exposure Durations*



4.3 Inferential Analysis (ANOVA Results)

To test whether gasoline exposure significantly affected biomarker expression across the different time intervals, a one-way Analysis of Variance (ANOVA) was conducted for each biomarker. This statistical test was chosen because it allows comparison of mean expression levels among more than two independent groups. The independent variable was *time of exposure* (Control, 1 hour, 3 hours, 5 hours, 7 hours, and 9 hours), while the dependent variable was *gene expression level*. The ANOVA analysis helped determine whether the observed changes in expression levels were statistically significant, indicating the presence of time-dependent variations in biomarker activity following gasoline exposure.

4.3.1 Interleukin-1 Beta (IL-1β)

Table 4.4 presents the results of the one-way ANOVA for Interleukin-1 Beta (IL-1β). The analysis revealed a significant difference in expression levels across the six time intervals, $F(5,6) = 9293.08$, $p < 0.001$. This indicates that the duration of gasoline exposure had a statistically significant impact on IL-1β expression. The post hoc Tukey HSD test further confirmed that the major source of variation occurred between the 1-hour exposure group and all other time groups, with the 1-hour group showing the highest mean expression (11.95 ± 0.11). This sharp elevation at 1 hour suggests an acute inflammatory response immediately after exposure, which is characteristic of IL-1β activation as an early-phase cytokine in inflammation.

In contrast, expression levels sharply decreased after the 1-hour mark and remained relatively stable from 3 to 9 hours, indicating a gradual normalization of IL-1β activity as the system adapted or compensated for the initial inflammatory insult. The subset grouping also confirmed that IL-1β expression at 5 hours formed a distinct cluster, separate from the highly elevated 1-hour group, further supporting a significant time-dependent modulation. The variation pattern demonstrates that gasoline exposure rapidly induces IL-1β production but the response diminishes with time, reflecting the transient nature of pro-inflammatory cytokine activity.

Table 4.4: One-Way ANOVA Result for IL-1β Expression Across Exposure Durations

Source of Variation	**Sum of Squares**	**df**	**Mean Square**	**F**	**Sig.**
Between Groups	202.512	5	40.502	9293.079	.000
Within Groups	0.026	6	0.004		
Total	**202.538**	**11**			

4.3.2 Tumor Necrosis Factor Alpha (TNF-α)

Table 4.5 shows the results of the one-way ANOVA test conducted to evaluate the effect of gasoline exposure duration on the expression of Tumor Necrosis Factor Alpha (TNF-α). The results indicate a statistically significant difference in TNF-α expression levels among the six exposure time groups, $F(5,6) = 20156.70$, $p < 0.001$. This suggests that exposure duration had a strong influence on TNF-α expression, revealing clear time-dependent changes following gasoline exposure. The Tukey HSD post hoc test further confirmed that the significant variations were largely driven by the 1-hour and 3-hour groups, which displayed remarkably higher mean expression levels (12.19 ± 0.05 and 13.50 ± 0.13, respectively) compared to the control and later exposure groups.

This pattern of TNF-α expression indicates a pronounced inflammatory surge within the first three hours of exposure, consistent with its role as a rapid-response cytokine in inflammatory signaling. The sharp increase at 1 and 3 hours reflects immediate activation of TNF-α following toxic insult, likely triggering other downstream inflammatory mediators. Expression levels subsequently declined sharply at 5 hours (0.39 ± 0.04) before returning to moderate levels between 7 and 9 hours (1.48 ± 0.04 and 1.22 ± 0.02, respectively), suggesting the onset of compensatory or recovery mechanisms. The subset groupings further support this trend, with the control, 5-hour, and 9-hour time points forming distinct homogeneous subsets separate from the acute response peaks at 1 and 3 hours. These findings confirm that gasoline exposure provoked a strong but transient TNF-α–mediated inflammatory response.

Table 4.5: One-Way ANOVA Result for TNF-α Expression Across Exposure Durations

Source of Variation	**Sum of Squares**	**df**	**Mean Square**	**F**	**Sig.**
Between Groups	375.419	5	75.084	20156.700	.000
Within Groups	0.022	6	0.004		
Total	**375.441**	**11**			

4.3.3 Caspase-3

The one-way ANOVA result for Caspase-3 expression across different exposure durations is presented in Table 4.6. The analysis revealed a statistically significant difference in mean expression among the groups, $F(5,6) = 135.68$, $p < 0.001$, indicating that exposure time had a significant effect on Caspase-3 activity. The post hoc Tukey HSD test confirmed that the significant variations were primarily between the 3-hour group and the later exposure durations (7 and 9 hours), as well as between the control and 1-hour groups. The highest mean expression was observed at 1 hour (1.29 ± 0.01), followed by 9 hours (1.28 ± 0.04) and 7 hours (1.13 ± 0.04), while the lowest was recorded at 3 hours (0.77 ± 0.02).

This pattern suggests that Caspase-3 activation was initially upregulated following exposure, possibly indicating early apoptotic signaling, then transiently suppressed at 3 hours before reactivation in the later hours. Since Caspase-3 is a key executioner enzyme in apoptosis, the significant time-dependent fluctuations highlight the dynamic cellular responses to gasoline exposure. The elevated expression at 1 and 9 hours implies that apoptotic processes may have been reinitiated after prolonged exposure, perhaps due to sustained oxidative or inflammatory stress. The homogeneous subsets in the post hoc test further confirmed that each time point represented a distinct phase of expression response, emphasizing that Caspase-3 activation followed a biphasic (two-peak) temporal pattern in this experiment.

Table 4.6: One-Way ANOVA Result for Caspase-3 Expression Across Exposure Durations

Source of Variation	**Sum of Squares**	**df**	**Mean Square**	**F**	**Sig.**
Between Groups	0.430	5	0.086	135.684	.000
Within Groups	0.004	6	0.001		
Total	**0.433**	**11**			

4.4 Discussion of Findings

The findings from this study revealed that exposure to gasoline fumes had a significant effect on the expression of Interleukin-1β (IL-1β) in the liver tissues. The ANOVA results showed a sharp rise in IL-1β levels at 1 hour post-exposure, followed by a gradual decline toward later hours. This indicates an early inflammatory response triggered by gasoline inhalation. This observation supports the report of Tilg et al. (2016) and Di Paolo and Shayakhmetov (2016), who explained that IL-1 family cytokines are rapidly synthesized in response to cellular stress and necrosis, leading to pro-inflammatory signaling cascades. The transient peak of IL-1β observed aligns with the mechanism described by Uboh et al. (2005), who found that gasoline exposure can provoke oxidative stress and enzyme elevation in the liver. The early upregulation of IL-1β in this study therefore suggests an acute inflammatory response resulting from toxic hydrocarbons in gasoline, which stimulates hepatocytes and macrophages to secrete cytokines that initiate inflammation.

Similarly, Tumour Necrosis Factor-alpha (TNF-α) expression showed a marked elevation after one hour of exposure, which was sustained up to the third hour before declining gradually. This trend indicates that TNF-α acts as a central mediator in the hepatic inflammatory response induced by gasoline fumes. The findings corroborate those of Bushra et al. (2014), who reported that gasoline inhalation can stimulate TNF-α release, resulting in hepatocellular inflammation and possible apoptosis. This agrees with Parameswaran and Patial (2010) and Bradley (2008), who described TNF-α as a multifunctional cytokine produced by macrophages and monocytes to regulate immune and inflammatory reactions. Furthermore, the increased TNF-α expression supports the report of Beggs et al. (2014) that exposure to petroleum derivatives induces TNF-α synthesis, which can damage liver cells through inflammatory and apoptotic pathways. The decline observed at later hours may reflect the downregulation of inflammatory signaling after the initial acute response, in

line with Brenner et al. (2013) who noted that prolonged cytokine activation can give way to compensatory mechanisms to restore hepatic homeostasis.

For Caspase-3, the expression analysis revealed a mild elevation after one hour of exposure, followed by fluctuations and eventual stabilization. This pattern suggests the initiation of apoptotic processes following prolonged exposure to gasoline fumes. The observed increase supports the findings of Elmore (2007) and Brown et al. (2014), who stated that Caspase-3 is a key executioner enzyme that mediates programmed cell death during toxic stress. Similarly, Owagboriaye et al. (2017) reported that gasoline exposure alters the ultrastructural integrity of liver cells, triggering apoptotic pathways through caspase activation. The activation of Caspase-3 observed in this study is also consistent with the mechanism described by Yang et al. (2014), who noted that apoptotic hepatocytes are characterized by intense Caspase-3 activity following TNF receptor stimulation. Hence, the current findings indicate that gasoline exposure not only induces inflammatory cytokines such as IL-1β and TNF-α but also activates Caspase-3-mediated apoptosis, suggesting a linked pathway between inflammation and programmed cell death in hepatic tissues.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

This study investigated the effect of gasoline fume inhalation on hepatic biomarkers — *Interleukin-1β (IL-1β)*, *Tumour Necrosis Factor-alpha (TNF-α)*, and *Caspase-3* — in experimental rats exposed for varying durations (Control, 1 hour, 3 hours, 5 hours, 7 hours, and 9 hours). The main objective was to determine how the duration of exposure to gasoline fumes influences inflammatory and apoptotic responses in liver tissues. Using one-way ANOVA analysis, significant differences ($p < 0.05$) were observed among all groups, confirming that exposure time had a notable impact on the expression levels of these biomarkers.

The descriptive statistics showed that IL-1β expression was lowest in the control group but increased markedly at 1 hour of exposure, representing an early inflammatory reaction. However, subsequent exposure for longer periods (3–9 hours) produced variable responses, suggesting a modulation of the inflammatory process as exposure progressed. This pattern aligns with the findings of Beggs et al. (2014) and Chen et al. (2017), who reported that gasoline fumes and other xenobiotics trigger an acute inflammatory response mediated by cytokines such as IL-1β. The initial surge and later decline imply that IL-1β acts as a first-line inflammatory signal, followed by the body's attempt to restore homeostasis.

Similarly, TNF-α expression significantly increased in the early hours of exposure, reflecting its role as an immediate pro-inflammatory cytokine in hepatic defense. The elevated TNF-α concentrations recorded in the exposed groups corroborate the reports of Bushra et al. (2014) and Uboh et al. (2005), which established that gasoline fumes can stimulate the overproduction of

inflammatory cytokines, leading to liver injury and oxidative stress. The gradual decrease observed after prolonged exposure suggests possible adaptation or inhibition of the inflammatory cascade, consistent with the regulatory feedback mechanisms described by Parameswaran and Patial (2010).

In contrast, Caspase-3 expression followed a different trend—initially low in the early exposure groups but progressively increased with longer exposure durations. This pattern signifies the transition from inflammation to apoptosis, as the sustained impact of gasoline fumes leads to programmed cell death. The result agrees with the studies of Yang et al. (2014) and Elmore (2007), who reported that activation of Caspase-3 occurs downstream of TNF-α signaling and serves as a biochemical marker of apoptosis in hepatic cells. The progressive activation of Caspase-3 in this study therefore indicates that prolonged exposure to gasoline fumes can cause irreversible hepatocellular damage.

Overall, the findings from this experiment confirm that gasoline fume inhalation disrupts hepatic stability by inducing both inflammatory and apoptotic pathways. The ANOVA results demonstrated statistically significant alterations in all measured biomarkers, confirming that time-dependent exposure plays a critical role in liver injury progression. These results support existing literature that described the hepatotoxic, inflammatory, and apoptotic consequences of gasoline exposure (Owagboriaye et al., 2017; Uboh et al., 2005; Bushra et al., 2014). The study concludes that prolonged inhalation of gasoline fumes poses a serious health risk by damaging liver tissue through cytokine-mediated inflammation and apoptosis.

5.2 Conclusion

This study has demonstrated that gasoline fume exposure significantly alters the expression of hepatic biomarkers — *Interleukin-1β (IL-1β)*, *Tumour Necrosis Factor-alpha (TNF-α)*, and *Caspase-3* — in a time-dependent manner. The findings revealed that short-term exposure induced an early inflammatory response through elevated IL-1β and TNF-α levels, while prolonged exposure led to increased Caspase-3 activity, signifying apoptotic cell death. These results confirm that gasoline fumes disrupt liver integrity by initiating inflammation and apoptosis, consistent with previous studies such as Uboh et al. (2005) and Bushra et al. (2014), which reported that petroleum vapours cause oxidative stress and hepatocellular damage.

The significant activation of TNF-α and IL-1β following exposure supports the position of Beggs et al. (2014) and Chen et al. (2017) that gasoline fumes act as potent stimulants of pro-inflammatory cytokines. Similarly, the marked upregulation of Caspase-3 in the later exposure periods corroborates the reports of Elmore (2007) and Yang et al. (2014) that apoptosis is a secondary outcome of prolonged chemical stress in hepatic tissues. Together, these observations underscore that persistent or repeated inhalation of gasoline fumes can trigger a chain of molecular events leading to liver injury and loss of cellular homeostasis.

In conclusion, this research confirms that gasoline fume inhalation poses a serious hepatotoxic threat, capable of inducing both inflammatory and apoptotic mechanisms through cytokine and caspase activation. The study provides further experimental evidence supporting the earlier assertions in literature that environmental and occupational exposure to petroleum products compromises hepatic function and structure. It therefore highlights the urgent need for stricter environmental controls, improved occupational safety measures, and public awareness on the health risks associated with prolonged gasoline fume exposure.

5.3 Recommendations

1. **Enforcement of Safety Regulations:** Government and relevant health authorities should ensure strict enforcement of occupational safety standards for workers exposed to gasoline fumes. Petrol attendants, refinery workers, and transporters should be mandated to use protective equipment such as masks and gloves during operation to minimize inhalation of toxic vapours.

2. **Environmental Pollution Control:** There is a need for continuous monitoring of environmental air quality around petrol stations and refineries to detect and control excessive gasoline emissions. Environmental agencies should also enforce penalties on individuals or organizations that dispose of petroleum products improperly, leading to soil or air contamination.

3. **Public Health Awareness:** Awareness programs should be organized regularly to educate the public and petroleum industry workers about the harmful effects of gasoline exposure. Such enlightenment should emphasize preventive practices, early detection of symptoms, and the importance of periodic medical examinations.

4. **Adoption of Clean and Sustainable Energy:** To reduce dependence on petroleum and its fumes, the use of cleaner energy sources such as solar, biogas, and wind energy should be promoted. Government incentives and subsidies could encourage households and industries to adopt eco-friendly energy alternatives.

5. **Encouragement of Further Research:** More detailed studies should be carried out to evaluate the biochemical and molecular effects of gasoline fumes on different organs. Future research

should also focus on identifying natural or synthetic agents that could mitigate oxidative stress and apoptosis caused by petroleum exposure.

5.4 Suggestions for Further Studies

1. Future studies should focus on investigating the long-term biochemical and histological effects of chronic gasoline fume exposure on various organs beyond the liver, such as the kidney, lungs, and brain, using larger animal models for broader validation.

2. Further research is needed to analyze gene expression patterns associated with apoptosis and inflammation-related biomarkers (such as *Caspase-3*, *TNF-α*, and *IL-1β*) to understand the molecular mechanisms underlying fuel-induced hepatic injury.

3. It is recommended that subsequent studies employ dose–response and recovery experiments to evaluate whether the hepatic and cellular changes observed are reversible after cessation of exposure.

4. Future work could compare the toxicity profiles of different petroleum derivatives (e.g., kerosene, diesel, and gasoline) to identify which product poses the greatest risk to liver function and apoptosis regulation.

5. Further investigations should be extended to human epidemiological studies among occupationally exposed populations (e.g., petrol station workers) to confirm the experimental findings and assess real-world health implications.

5.5 Limitations of the Study

1. The study was conducted using a small sample size, which may limit the generalizability of the findings to larger populations or different species.

2. Only acute exposure durations were considered, hence the effects of long-term or chronic gasoline fume inhalation were not evaluated.

3. The study focused mainly on biochemical and molecular biomarkers (such as TNF-α and Caspase-3) without detailed histopathological examinations of liver tissues that could further validate cellular damage.

4. Environmental variables such as temperature, humidity, and ventilation conditions during exposure were not controlled or standardized, which might have influenced the concentration of inhaled fumes.

5. Due to limited resources, the study did not include comparative analysis with other petroleum products or antioxidant treatment groups, which could provide insight into possible protective or mitigating interventions.